

Problem Statement Canvas 1

1. User / Protagonist:

We had 4 personas, but the most repeated one is of the:

Persona: Conflicted Power User (Student)

Description:

18–22-year-old undergraduate student who frequently uses AI tools like ChatGPT for assignments, studying, and productivity. High tech comfort, relies on AI especially under time pressure, but feels uncertain about correctness and ethical usage.

2. Context & Situation:

When/Where:

- During assignments, exams, project work
- Late-night deadlines / time pressure situations
- Academic environments with unclear AI policies

Trigger:

- Need to complete work quickly
- Difficulty understanding concepts
- Peer usage of AI

Environment:

- University setting
- Digital learning environments

Secondary Stakeholders:

- Teachers / professors
- Institutions (universities)
- Employers (future impact)

3. Evidence Base:

Methods Used:

- Semi-structured interviews
- Conversational field interviews (road-style)
- Persona synthesis

Key Data Points (REAL insights):

- “I don’t know where the line is between help and cheating”
- “I use it almost daily”
- “If it’s important, I cross-check... otherwise I just use it”
- “It saves time but I’m not fully sure if it’s correct”

4. Pain Points & Needs:

Frustrations:

- No clear boundary between *help* vs *cheating*
- Cannot fully trust AI outputs
- Inconsistent decision-making
- Fear of over-dependence

Unmet Needs:

- Clear guidance on how to use AI
- Confidence in decision-making
- Understanding when to trust AI

5. Root Causes (5 Whys)

- Unclear boundaries of AI usage
- Lack of contextual ethical guidance
- AI systems are output-focused, not understanding-focused
- Success metrics prioritize capability over user comprehension
- AI operates as a black box with limited transparency

Summarised: Lack of **trust infrastructure** — users receive powerful outputs without guidance, transparency, or guardrails to use them responsibly.

6. Impact & Scale:

Severity: High

- Affects decision-making and learning quality

Frequency: Very frequent

- Daily/weekly usage among students

Scale:

- Majority of students using AI tools

Consequences if unresolved:

- Reduced learning depth
- Ethical misuse
- Over-dependence on AI
- Anxiety and confusion

Current Workarounds:

- Cross-checking manually
- Editing AI responses
- Avoiding AI in “important” tasks

Why they fail:

- Inconsistent
- Time-consuming
- Still doesn’t remove uncertainty

7. POV Statement

Students who frequently use AI need clarity on how to use it responsibly because they experience confusion and uncertainty about ethical boundaries, leading to inconsistent and unreliable decision-making.

8. How Might We (HMW)

- How might we help students understand where to draw the line between AI assistance and misuse?
- How might we provide contextual guidance on when and how to use AI responsibly?
- How might we help users decide when to trust AI outputs?
- How might we reduce confusion and anxiety around AI usage?

9. Reframe Check

The problem statement needs change:

Initially we thought the issue was lack of clear guidelines, but from our interviews we realized that users don’t necessarily follow guidelines even when they exist. Their

behavior is driven more by convenience and perceived usefulness. So we're shifting our focus toward awareness, trust, and decision-making while using AI.

Problem statement revised

1. User / Protagonist

Persona: Conflicted Power User (Student)

Description:

18–22-year-old undergraduate student who frequently uses AI tools for assignments, studying, and productivity. Highly tech-comfortable, relies on AI under time pressure, prioritizes convenience, but lacks clear understanding of AI limitations and risks.

2. Context & Situation

When/Where:

- During assignments, exams, project work
- Especially under deadlines or time pressure
- In academic environments with increasing AI usage

Trigger:

- Need to complete work quickly
- Difficulty understanding topics
- Peer influence (others using AI)

Environment:

- Digital learning + personal devices
- Minimal supervision while using AI

Secondary Stakeholders:

- Professors / institutions
- Employers (long-term skill impact)
- AI platforms

3. Evidence Base

Methods Used:

- Semi-structured interviews
- Conversational (open-ended) field interviews
- Persona synthesis

Key Data Points:

- “I use it almost daily”
- “If it’s important, I cross-check... otherwise I just use it”
- “I don’t know where the line is between help and cheating”
- “It saves time but I’m not sure if it’s always correct”

Observation: Users continue using AI **despite uncertainty**

4. Pain Points & Needs

Frustrations:

- Cannot judge when to trust AI
- Lack of awareness of limitations
- Inconsistent verification behavior
- Over-reliance without understanding

Unmet Needs:

- Ability to make **better decisions while using AI**
- Awareness of risks and limitations
- Confidence in when/how to rely on AI

5. Root Causes (5 Whys)

- Users rely on AI due to convenience
- They lack awareness of limitations and risks
- AI systems provide outputs without contextual guidance
- Systems are designed for performance, not user understanding
- Transparency and interpretability are limited in AI design

Deep Root Cause:

Lack of **trust calibration and awareness infrastructure** — users are not equipped to judge when to trust, question, or verify AI outputs.

6. Impact & Scale

Severity: High

- Affects learning quality and decision-making

Frequency: Very high

- Daily/weekly usage among students

Scale:

- Majority of students using AI tools

Consequences if unresolved:

- Misplaced trust in incorrect outputs
- Reduced critical thinking
- Over-dependence on AI
- Poor decision-making

Current Workarounds:

- Occasional cross-checking
- Editing AI responses
- Avoiding AI in important cases

Why they fail:

- Inconsistent
- Based on guesswork
- Do not build true understanding

7. Point of View (POV Statement)

Students who frequently use AI need to make better decisions about when and how to rely on it because they lack awareness of its limitations and cannot accurately judge its trustworthiness.

8. How Might We (HMW)

- How might we help students decide when to trust AI outputs?
- How might we make users more aware of AI limitations without disrupting convenience?
- How might we reduce over-reliance while still preserving productivity?
- How might we help users verify AI responses more effectively?
- How might we design AI interfaces that visually communicate certainty levels or source citations, especially for complex technical tasks like debugging or system architecture?
- How might we create "productive friction" in the AI experience that slows users down just enough to engage their critical thinking before copying text?
- How might we prompt students to reflect on and summarize the AI's logic in their own words to ensure comprehension?

- How might we shift the user's perception of AI from an infallible "answer engine" to a collaborative "sounding board"?
- How might we seamlessly integrate rapid, independent fact-checking tools directly into the student's existing digital learning environment?
- How might we help students track their own independent learning progress separately from the AI's generated outputs?
- How might we reward users for independently verifying and correcting AI-generated claims?

9. Reframe Check

Alternative framing:

- What if the problem is not misuse, but **misplaced trust**?
- What if users don't need rules, but **better judgment tools**?

Who might see this as a feature?

- Power users who benefit from speed over accuracy
- Companies optimizing for engagement and usage

10. Final Problem Statement

Conflicted student users experience misplaced trust, inconsistent verification, and over-reliance on AI because they lack awareness of its limitations and the ability to judge its reliability.

How might we help them make better decisions while using AI so that they can maintain productivity while improving accuracy, confidence, and responsible usage?

FINAL EVALUATION:

Your Problem Statement

Undergraduate students using AI for academic tasks experience uncertainty, misplaced trust, and inconsistent verification because they lack awareness of AI limitations and the ability to judge its reliability in context. How might we help them make better decisions while using AI so that they can improve accuracy, reduce blind reliance, and demonstrate more consistent verification behavior in their academic work?

Dimension 1 — Human-Centeredness

1.1 Named or specific protagonist?

Answer: Yes

Reasoning: The statement clearly identifies a specific user group: undergraduate students using AI for academic tasks. It is not vague like “people” or “users.”

1.2 Grounded in empathy research?

Answer: Yes

Reasoning: The statement comes from your interviews and persona synthesis. Your research repeatedly showed students using AI frequently, cross-checking inconsistently, and feeling unsure about trust and responsible use.

1.3 Emotional context present?

Answer: Partial

Reasoning: The words uncertainty and misplaced trust imply emotional difficulty, but the statement does not directly mention feelings like anxiety, guilt, or confusion. It is emotionally informed, but not strongly emotional in wording.

Dimension 2 — Solution-Agnosticism

2.1 No technology or method prescribed?

Answer: Yes

Reasoning: The problem statement does not assume a specific solution like an app, workshop, policy, or tool. It stays at the level of the user need.

2.2 Passes the “elevator speed” reframe test?

Answer: Yes

Reasoning: The statement can still be reframed in different ways, such as awareness, trust calibration, critical thinking, or misuse prevention. It does not lock you into one single path.

2.3 Free from the “Law of the Hammer” bias?

Answer: Yes

Reasoning: The statement is not framed around something your team wants to build. It focuses on the user’s behavior and decision-making problem rather than a preferred solution.

Dimension 3 — Frame Quality

3.1 Causal chain articulated?

Answer: Yes

Reasoning: The statement clearly explains the cause using because: students experience these problems because they lack awareness of AI limitations and the ability to judge reliability.

3.2 Context and constraints visible?

Answer: Yes

Reasoning: The context is visible: undergraduate students, AI, and academic tasks. This tells us when and where the problem occurs.

3.3 Avoids premature narrowing?

Answer: Yes

Reasoning: The statement is specific enough to act on, but still broad enough for multiple interpretations and solutions. It does not prematurely reduce the problem to just ethics, cheating, or policy.

Dimension 4 — Measurability

4.1 Quantitative success criteria?

Answer: Partial

Reasoning: It includes outcomes like improve accuracy, reduce blind reliance, and more consistent verification, which are measurable in principle. But it does not yet include exact numbers or percentages.

4.2 Observable and verifiable?

Answer: Yes

Reasoning: A neutral observer could check whether students verify outputs more often, rely less blindly, and make fewer incorrect trust decisions. These are observable behaviors.

4.3 Time-bound or scoped?

Answer: No

Reasoning: The statement does not specify a time frame such as “within one semester” or “during assignments over two weeks.” It is scoped to academic work, but not time-bound.

Dimension 5 — Scope Calibration

5.1 Broad enough for multiple solutions?

Answer: Yes

Reasoning: The problem could be addressed through many directions: decision aids, awareness tools, learning interventions, interface changes, peer systems, or educator support.

5.2 Narrow enough to be actionable?

Answer: Yes

Reasoning: It is focused on one main user group and one situation: students using AI for academic tasks. That is manageable for a student project.

Dimension 6 — Ethical & Systemic Awareness

6.1 Stakeholder awareness?

Answer: Partial

Reasoning: The statement centers students, but other stakeholders like teachers, institutions, and future employers are only indirectly implied, not explicitly mentioned.

6.2 Free from harmful assumptions?

Answer: Yes

Reasoning: It does not stereotype students as lazy or dishonest. It frames them as users facing a real judgment and awareness problem, which is fair and balanced.

Reflective Notes

The statement is strong because it is user-centered, research-backed, and clearly explains the causal chain. Its weakest area is measurability, because it lacks exact targets and time scope. It could be improved further by adding a measurable success condition such as increasing verification behavior or reducing blind trust by a certain percentage.

Alternative Reframings

- What if the problem is not unethical use, but misplaced trust in AI outputs?
- What if the problem is not lack of rules, but lack of judgment while using AI?
- What if the issue is not AI itself, but the gap between convenience and critical thinking?